Homework # 4 : Ch 7+8 EGCI 230 : Electric Circuit Analysis

Submitted by…………………………

1. For the circuit in Fig. 1, find :
   a) $i(0+)$ and $v(0+)$.
   b) $di(0+)/dt$ and $dv(0+)/dt$.
   c) $i(\infty)$ and $v(\infty)$.

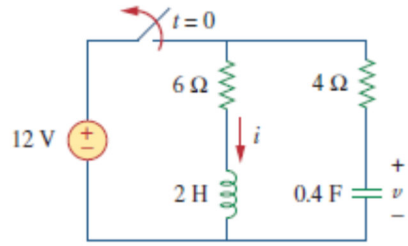
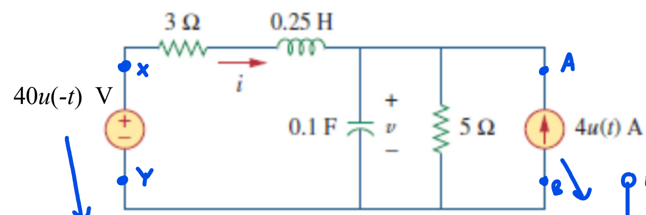
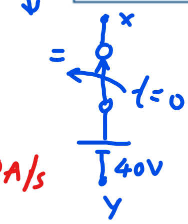


Fig. 1.

Ans
a) $i(0^+) = 2A$; $v(0^+) = 12V$
b) $\frac{di(0^+)}{dt} = -4A/s$; $\frac{dv(0^+)}{dt} = -5V/s$
c) $i(\infty) = 0A$; $v(\infty) = 0V$

2 For the circuit in Fig. 2, find :
   a) $v(0+)$ and $i(0+)$.
   b) $dv(0+)/dt$ and $di(0+)/dt$.
   c) $v(\infty)$ and $i(\infty)$.





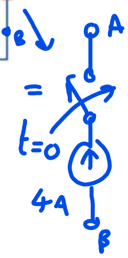


Fig. 2.

Ans
a) $v(0^+) = 25V$; $i(0^+) = 5A$
b) $\frac{dv(0^+)}{dt} = 40V/s$; $\frac{di(0^+)}{dt} = -160A/s$

3. Obtain expressions for the current $i(t)$ and voltage $v(t)$ as labeled in the circuit of Fig. 3. which are valid for all $t > 0$.

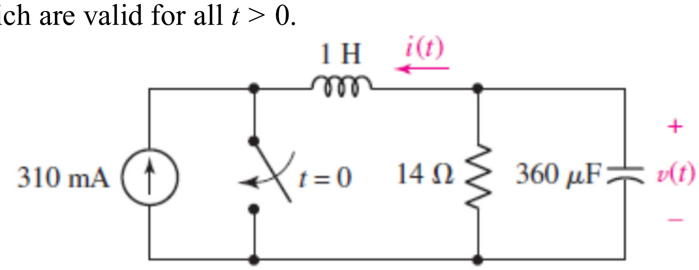
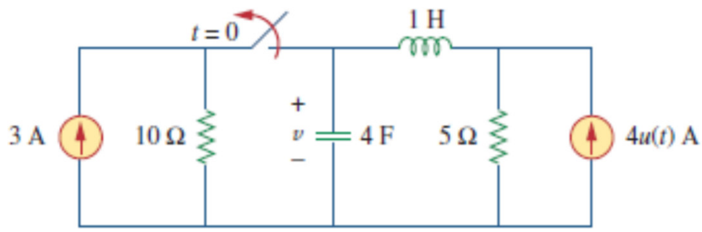


Fig. 3.

Ans This problem is the 2nd-order → skip But $i_L(0^-) = i_L(0^+) = -310\,mA$
$v_C(0^-) = v_C(0^+) = 4.34\,V$

Submitted by…………………………

4. Find $v(t)$ for $t > 0$ in the circuit of Fig. 4.



2nd-order "skip"

But $i_L(0) = 2A$; $V_C(0) = 10\ V$

Fig. 4.

5. Obtain $v(t)$ and $i(t)$ for $t > 0$ in the circuit of Fig. 5.

2nd-order : skip

But

$V(0) = 20\ V$

$i(0) = 0A$

$\frac{dv(0)}{dt} = 0\ V/s$

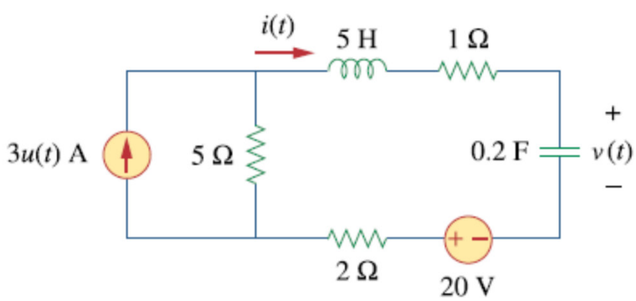
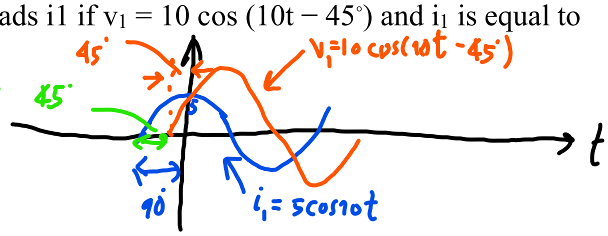


Fig. 5.

6. Determine the angle by which v_1 leads i1 if $v_1 = 10\cos(10t - 45^\circ)$ and i_1 is equal to

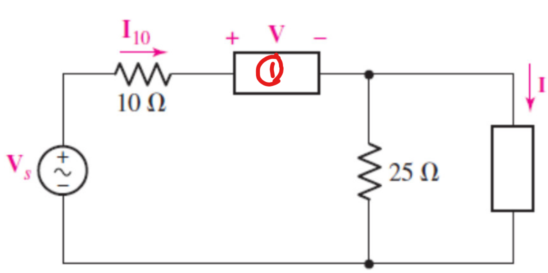
(a) $5\cos 10t$ = 45°
(b) $5\cos(10t - 80^\circ)$
(c) $5\cos(10t - 40^\circ)$
(d) $5\cos(10t + 40^\circ)$
(e) $5\sin(10t - 19^\circ)$.

a) only



7. In the circuit of Fig.7 , which is shown in the phasor (frequency) domain, $\mathbf{I}_{10}$ is determined to be $2\angle 42^\circ$ mA. If $\mathbf{V} = 40\angle 132^\circ$ mV:

(a) what is the likely type of element connected to the right of the 10 Ω resistor and
(b) what is its value, assuming the voltage source operates at a frequency of 1000 rad/s?



Ans

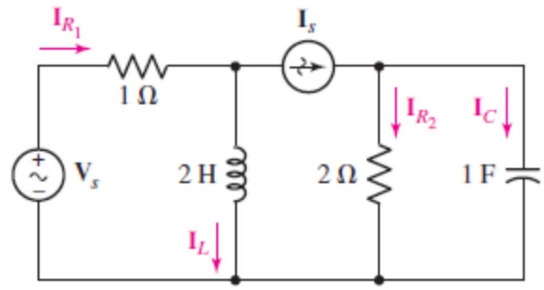
a) ① is L

b) L = 20 mH

Fig. 7.

Submitted by…………………………

8. For the *RLC* circuit of Fig. 7, determine $\mathbf{I_s}$ and $i_s(t)$ if both sources operate at $\omega = 2$ rad/s, and $\mathbf{I_C} = 2\angle 28^\circ$ A.



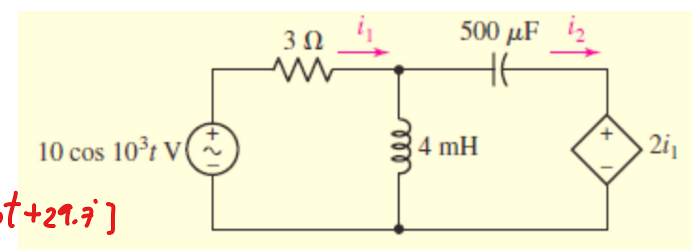
Ans

$I_s = 1.5\angle -62^\circ$

$i_s(t) = 1.5\cos(2t - 62^\circ)$

Fig. 8.

9. Obtain expressions for the time - domain currents $i_1(t)$ and $i_2(t)$ in the circuit given as Fig. 8.



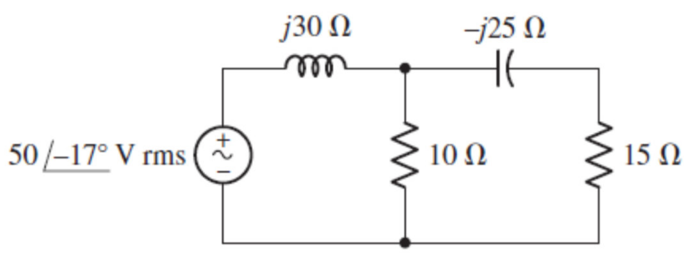
Ans

$i_1(t) = 1.24\cos[1000t + 29.7^\circ]$

$i_2(t) = 2.77\cos[1000t + 56.3^\circ]$

Fig. 9.

10. Calculate the complex power delivered to each passive component of the circuit shown in Fig. 10, and determine the power factor of the source.



Ans complex power

Fig. 10.

$S_{j30\Omega} = 88.40\angle 90^\circ$ VA (OR $Q = 88.40$ VAR)

$S_{10\Omega} = 20\angle 0^\circ$ VA (OR $P_{10\Omega} = 20$ Watt)

$S_{-j25\Omega} = 5.865\angle -90^\circ$ VA (OR $Q = -5.865$ VAR)

$S_{15\Omega} = 3.517\angle 0^\circ$ VA (OR $P_{15\Omega} = 3.517$ Watt)

$S_{source} = 85.85\angle 74.05^\circ$ VA
$= 23.59 + j82.54$ VA

p.f $= \cos(74.05^\circ)$
$= 0.274$ (lagging)